Assignment 1: Student Grade Management and File Upload System

Course: PHP Web Development
Chapters: Chapter 3 (PHP Functions) and Chapter 4 (Forms, Validation and File Uploading)
Duration: 2 Hours and 30 Minutes
Total Marks: 5
Assignment Type: Individual Practical Assignment

Assignment Objective

Develop a Student Grade Management and File Upload System using HTML and PHP.

The system must allow users to enter student information, upload a certificate, calculate examination results using PHP functions, and display the results in an HTML table.

Task 1: Create Student Information Form

Create an HTML file named index.html containing the following fields:

Field	Input Type
Student Full Name	Text
Student ID	Text
Gender	Radio Button
Faculty	Drop-down List
PHP Mark	Number
Database Mark	Number
Networking Mark	Number
Student Certificate	File Upload

Requirements:

1. Use the POST method to submit information.
2. Use **enctype="multipart/form-data"** to support file uploading.
3. Add a **Submit and Calculate** button.
4. Add a **Reset** button.
5. Send the form information to **process.php**

Task 2: Create PHP Functions

Create a PHP file named functions.php.

Define the following four functions:

1. calculateTotal() to calculate the total marks of three subjects.
2. calculateAverage() to calculate the student's average marks.
3. calculateGrade() to determine the student's grade.
4. checkResult() to determine whether the student passed or failed.

Use the following grading criteria:

Average Marks	Grade
90–100	A
80–89.99	B
70–79.99	C
60–69.99	D
Below 60	F

A student passes when the average is 60 or above.

Each function must accept the necessary parameters and return its result.

Task 3: Process and Validate Form Information

Create a PHP file named process.php.

Use $_POST to retrieve the submitted student information.

Apply the following validation rules:

1. Student Name must not be empty.
2. Student ID must not be empty.
3. Gender must be selected.
4. Faculty must be selected.
5. All three subject marks must be entered.
6. Each subject mark must be between 0 and 100.

Display an appropriate error message when validation fails.

Use PHP conditional statements, isset(), and empty() where appropriate.

Task 4: Upload Student Certificate

Create a folder named uploads to store student certificates.

Implement the following features:

1. Allow the user to upload one certificate.
2. Retrieve the uploaded file information using $_FILES.
3. Check whether a file was successfully submitted.
4. Reject files larger than 2 MB.
5. Check whether a file with the same name already exists using file_exists().
6. Save the uploaded certificate in the uploads folder using move_uploaded_file().
7. Display an appropriate message when uploading succeeds or fails.
8. Do not overwrite an existing file.

Task 5: Display Student Results (15 Marks)

After successful validation and file uploading, display the student information and examination results in an HTML table.

The table must contain:

1. Student Full Name
2. Student ID
3. Gender
4. Faculty
5. PHP Mark
6. Database Mark
7. Networking Mark
8. Total Marks
9. Average Marks

10. Grade
11. Pass or Fail
12. Certificate Name
13. File Type
14. File Size
15. Upload Status

Use **require_once** to include the functions defined in **functions.php.**

General Instructions

1. Complete the assignment individually.
2. Use HTML and PHP.
3. Do not use JavaScript, a database, or frameworks.
4. Do not create arrays or use array-processing loops.
5. Use the PHP concepts covered in Chapters 3 and 4.
6. Include comments explaining important sections of your code.
7. Keep the HTML form and PHP processing code in separate files.
8. Test the application using XAMPP.
9. Ensure that the form, calculations, validation, and file uploading work correctly.

Required Project Structure

Student_Grade_System/

├── index.html

├── functions.php

├── process.php

└── uploads/

Submission Requirements

Submit:

1. The complete **Student_Grade_System** folder.
2. All three-source code files.
3. The uploads folder contains one sample certificate.
4. A screenshot of the student information form.
5. A screenshot showing the examination results in an HTML table.
6. A screenshot showing a successfully uploaded certificate.
7. A screenshot showing at least one validation error.

Marking Scheme

Assessment Criteria	Marks
HTML Student Information Form	20
PHP Functions	20
From Processing and Validation	20
Certificate Uploading	25
Displaying Student Results	15
Total	**100**